Abdelrahman Khalifa

Software Engineer — Rust · C++ · Linux · GSoC 2026 · Codeforces Expert

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SUMMARY

Software engineer working in Rust, modern C++, C and Python. Completed Google Summer of Code 2026 with The Linux Foundation (7 PRs merged upstream) and sole software engineer at EKSON, owning firmware, desktop app and backend. Codeforces Expert, 4-time ECPC finalist.

TECHNICAL SKILLS

Languages: Rust, C++ (modern), C, Python, JavaScript (Node.js), SQL, Java, C#, Bash

Computer science: Data structures and algorithms, operating systems, concurrency and multithreading, async/event-driven design, memory management, networking (TCP/UDP, IPC)

Systems & Linux: Linux (Ubuntu/Debian), FFI, IPC (varlink), CUPS/IPP, DNS-SD/mDNS, kernel modules and device drivers, Zephyr RTOS, FreeRTOS, Yocto, Buildroot

Backend & tools: Node.js, Express, MongoDB, REST APIs, PyQt5; Git/GitHub, GitHub Actions CI, Cargo, CMake, testing, debugging, code review

OPEN SOURCE

Google Summer of Code 2026 Contributor — The Linux Foundation / OpenPrinting | May 2026 – Oct 2026

COSMIC desktop printer setup tool in Rust; passed the final evaluation, still maintaining it | github.com/Abd002/GSOC-2026

- Designed a printer-management stack for the COSMIC desktop as 5 Rust crates: shared types, an async client, a CUPS/IPP/DNS-SD backend, a shared UI and a standalone app.
- Ran the C FFI backend as a varlink IPC service in the settings daemon, so a native crash cannot take down Settings.
- **7 PRs merged upstream**: ported cups-rs to libcups3 with CUPS 2 and 3 builds and CI, added DNS-SD bindings, and fixed crashes and a leaked client in libcups (C).
- Tested on real HP printers; CI runs formatting, Clippy and tests on several architectures with virtual IPP printers.

Contributor — Zephyr RTOS, Boa JavaScript engine (Rust), Rust Clippy | Mar 2025 – Dec 2025

- Zephyr RTOS: added the `IP_MULTICAST_LOOP` socket option for IPv4 across 6 networking modules (#86977, merged).
- Boa: optimized `JsStr::code_points()` for Latin1 strings, avoiding UTF-16 conversion (PR #4553, merged).
- Rust Clippy: fixed `ref_as_ptr` false positives for non-temporary references in `let/const` (PR #16214, merged).

EXPERIENCE

Embedded Software Engineer — EKSON | Aug 2025 – Present

Sole software engineer for 3-DoF and 6-DoF pneumatic motion simulators: firmware, desktop app, backend and CI

- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS: PID valve control with IMU feedback, safety stops and a UART protocol.
- Built the PyQt5 desktop control app and cut its CPU usage and latency by **~40%** (measured) by moving serial I/O off the UI thread.
- Built remote operation and firmware updates: a Node.js/Express + MongoDB backend that the desktop app polls over REST to relay commands and flash new firmware with esptool; GitHub Actions CI.

Embedded Linux Systems Intern — Orange Digital Center (ODC) | Dec 2023 – Jan 2024

- Built Linux images with Yocto and Buildroot and wrote a Linux kernel module; worked with device trees, root file systems and hardware flashing.

PROJECTS

- **IoT Device Communication System** (C++, Raspberry Pi 4, QEMU, Yocto): object-oriented framework for TCP unicast and UDP multicast between a server and emulated clients. GitHub
- **Operating System Simulator** (Java): CLI simulating scheduling, memory management, system calls and locks. GitHub
- **Inventory Management System** (C#, WinForms, SQL): desktop app for stock, orders and deliveries with role-based access control and automated reports. GitHub
- **Team projects (co-developed):** Sarh Al-Tafawuq, a production web platform (live); Nafaqati, a Flutter Android expense tracker (Google Play).

ACHIEVEMENTS

- Codeforces Expert (handle _Khalifa); 4-time ECPC finalist, 2021–2024 (Egyptian Collegiate Programming Contest).
- IEEE student branch Treasurer: managed a budget of more than $10k.

EDUCATION

B.Sc. Computer and Systems Engineering — Minia University | 2020 – 2025

Very Good (84.6%) with Honors. Graduation project: AUTOSAR-based V2X vehicle on STM32; I built the firmware update path and a dual-bank bootloader with rollback.